PRACTICAL – 4

Implementation and Time analysis of factorial program using iterative and recursive method

Aim:
To implement and analyze the time complexity of a factorial program using both iterative and recursive methods, and compare their performance.

Objective:
The objective of this experiment is to implement a factorial program using both iterative and recursive methods and analyze their performance. The experiment aims to understand the working of iteration and recursion, determine the time and space complexity of both approaches, and compare their efficiency in calculating the factorial of a given number.

1.Iterative Method

Algorithm:

- Start.
- Read n.
- Set fact = 1.
- Repeat from i = 1 to n:
- fact = fact * i
- Display fact.
- Stop.

Code:

```
# Iterative Factorial Function
def factorial_iterative(n):
    fact = 1

    for i in range(1, n + 1):
        fact = fact * i
```

```
    return fact


# Take input
n = int(input("Enter a number: "))

# Calculate factorial
result = factorial_iterative(n)

# Display result
print("Factorial using Iterative Method:", result)

# Enter your EN.No
print("EN.No:92460118698")
```

Output:

```
PS C:\Users\manvi\OneDrive\Desktop\daa practical 4> & C:\Users\manvi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/manvi/OneDrive/Desktop/daa practical 4/Factorial.py"
Enter a number: 5

Factorial using Iterative Method: 120
Factorial using Recursive Method: 120

Time Analysis:
Iterative Method: O(n)
Recursive Method: O(n)

Enrollment Number: 92460118698
```

Time Complexity:

Case	*Time Complexity*	*Description*
Best Case	*O(n)*	*Performs n iterations.*
Average Case	*O(n)*	*Performs n iterations.*
Worst case	*O(n)*	*Performs n iterations.*

Space Complexity:

Space Complexity	*Description*
O(1)	*Uses constant extra space.*

2.Recursive Method

Algorithm:

- Start.
- Read n.
- If n = 0 or n = 1, return 1.
- Otherwise, return n × factorial(n-1).
- Display the result.
- Stop.

Code:

```
# Recursive Factorial Function
def factorial_recursive(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial_recursive(n - 1)


# Take input
n = int(input("Enter a number: "))

# Calculate factorial
result = factorial_recursive(n)

# Display result
print("Factorial using Recursive Method:", result)

# Enter your EN.No
print("EN.No:92460118698")
```

Output:

```
PS C:\Users\manvi\OneDrive\Desktop\daa practical 4> & C:\Users\manvi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/manvi/OneDrive/Desktop/daa practical 4/Factorial.py"
Enter a number: 5

Factorial using Iterative Method: 120
Factorial using Recursive Method: 120

Time Analysis:
Iterative Method: O(n)
Recursive Method: O(n)

Enrollment Number: 92460118698
```

Time Complexity:

Case	*Time Complexity*	*Description*
Best Case	*O(n)*	*Makes n recursive calls.*
Average Case	*O(n)*	*Makes n recursive calls*
Worst case	*O(n)*	*Makes n recursive calls*

Space Complexity:

Space Complexity	*Description*
O(n)	*Uses recursion stack for n function calls.*